CHAPTER

Taylor's and Maclaurin's Series

Introduction :

- Taylor series is a representation of a function as an infinite sum of terms (i.e. series) that are calculated from the values of the functions derivative at a single point.

$$f(x+h) = f(x) + h\cdot f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots + \frac{h^n}{n!}f^n(x) + \dots$$

- Putting $x=a$ & $h=x-a$, we get Taylor's series in powers of $(x-a)$ as

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^n}{n!}f^n(a) + \dots$$

- Maclaurin's series,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^n(0) + \dots$$

- Basically, this section shows how infinitely differentiable function generates power series.

Taylor's series:

statement:

If $f(x+h)$ is a given function of h which can be expanded into positive integral powers of h, then

$$f(x+h) = f(x) + h\cdot f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots + \frac{h^n}{n!}f^n(x) + \dots$$

This is known as Taylor's series.

Putting $x=a$ & $h=x-a$ in above series, we get Taylor's series in powers of $(x-a)$ as

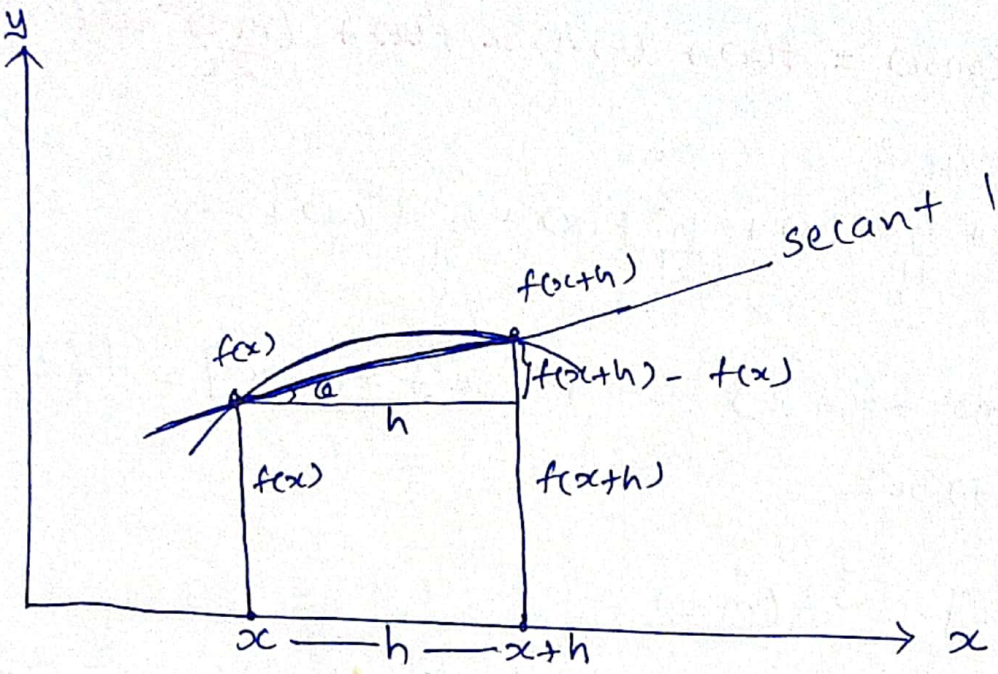
$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots$$
$$(x-a)^n/n!\ f^n(a) + \dots$$

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Note: To express the function in ascending powers of x, express h in terms of x.

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Introduction



$$f(x+h) = f(x) + \frac{h}{1!} \cdot f'(x) + \frac{h^2}{2!} f''(x) + \text{--- H.O.T.}$$

$$f'(x) = \frac{f(x+h) - f(x)}{h}$$

$$\frac{df}{dx} = \frac{f(x+h) - f(x)}{h}$$

To find slope of secant line Taylor series is a mapping function. It maps the curve.

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Examples:

① Prove that $f(mx) = f(x) + (m-1)x f'(x) + \frac{(m-1)^2 x^2}{2!} f''(x) + \ldots$

By Taylor's series

$$f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \ldots$$

$f(mx) = f(mx - x + x) = f[x + mx - x] = f\{x + (m$

$\therefore h = (m-1)x$

$\therefore f[x + (m-1)x] = f(mx)$

$$= f(x) + \frac{(m-1)x}{1!} f'(x) + \frac{(m-1)^2 x^2}{2!} f''(x$$

② Express $7 + (x+2) + 3(x+2)^3 + (x+2)^4 - (x+2)^5$ in ascending powers of x.

Let, $f(x+2) = 7 + (x+2) + 3(x+2)^3 + (x+2)^4 - (x+2)^5$

$f(x) = 7 + x + 3x^3 + x^4 - x^5$

By Taylor's series

$$f(x+h) = f(x) + h \cdot f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \frac{h^4}{4!} f^{iv}(x$$

Putting $h = 2$

$$f(x+2) = f(x) + 2 \cdot f'(x) + \frac{(2)^2}{2!} f''(x) + \frac{(2)^3}{3!} f'''(x) + \frac{(2)^4}{4!} f$$

$$= 7 + x + 3x^3 + x^4 - x^5 + 2[1 + 9x + 4x^3 - 5x^4]$$
$$\frac{4}{2}[18 + 12x^2 - 20x^3] + \frac{8}{6}[24$$

$f'(x) = 1 + 9x^2 + 4x^3 - 5x^4$, $f''(x) = 18x + 12x^2 - 20$

$f'''(x) = 18 + 24x - 60x^2$, $f^{iv}(x) = 24 - 120x$

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$$\therefore f(x+2) = (7+x+3x^3+x^4-x^5) + 2(1+9x^2+4x^3-5x^4) + \frac{2^2}{2!}(18x+12x^2-20x^3) + \frac{2^3}{3!}(18+24x-60x^2) + \frac{2^4}{4!}(24-120x) + \frac{2^5}{5!}(-120) + \cdots$$

$$= 7+x+3x^3+x^4-x^5+2+18x^2+8x^3-10x^4 + 2(18x+12x^2-20x^3) + \frac{4}{3}(18+24x-60x^2) + \frac{2}{3}(24-120x) + \frac{4}{15}(-120)$$

$$= 7+x+3x^3+x^4-x^5+2+18x^2+8x^3-10x^4 + 36x+24x^2-40x^3+24+32x-80x^2 + 16-80x-32$$

$$= 17-11x-38x^2-29x^3-9x^4-x^5+\cdots$$

Express $2x^3+7x^2+x-6$ in ascending powers of $(x-2)$

By Taylor's series,

$$f(x) = f(a) + (x-a)\cdot f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots$$

$f(x) = 2x^3+7x^2+x-6$

Putting $a=2$,

$$f(x) = f(2) + (x-2)f'(2) + \frac{(x-2)^2}{2!}f''(2) + \frac{(x-2)^3}{3!}f'''(2) + \cdots \quad \text{—(1)}$$

$f(x) = 2x^3+7x^2+x-6$, $f(2) = 2(2)^3+7(2)^2+2-6 = 16+28-4 = 40$

$f'(x) = 6x^2+14x+1$, $f'(2) = 6(2)^2+14(2)+1 = 53$

$f''(x) = 12x+14$, $f''(2) = 12(2)+14 = 38$

$f'''(x) = 12$, $f'''(2) = 12$ and so on.

$\therefore$ From eq^n (1),

$$f(x) = 40 + (x-2)(53) + \frac{(x-2)^2}{2!}(38) + \frac{(x-2)^3}{3!}(12) + \cdots$$

$$= 40 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3 + \cdots$$

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(4) Expand $\log \sin x$ in powers of $(x-2)$.

By Taylor's series,

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \ldots$$

Here, $f(x) = \log \sin x$

Putting $a = 2$,

$$f(x) = f(2) + (x-2) f'(2) + \frac{(x-2)^2}{2!} f''(2) + \frac{(x-2)^3}{3!} f'''(2) \quad \text{---}$$

Now,

$f(x) = \log \sin x$, $\quad f(2) = \log \sin 2$

$f'(x) = \frac{\cos x}{\sin x} = \cot x$, $\quad f'(2) = \cot 2$

$f''(x) = -\operatorname{cosec}^2 x$, $\quad f''(2) = -\operatorname{cosec}^2 2$.

$f'''(x) = -2 \operatorname{cosec}^2 x \cdot \cot x$, $\quad f'''(2) = 2 \operatorname{cosec}^2(2) \cdot \cot(2)$ an

From eqⁿ (1),

$$f(x) = \log \sin 2 + (x-2) \cdot \cot 2 + \frac{(x-2)^2}{2!} [-\operatorname{cosec}^2 2]$$

$$\frac{(x-2)^3}{3!} [2 \operatorname{cosec}^2(2) \cdot \cot(2)] + \ldots$$

$$\therefore \log \sin x = \log \sin 2 + (x-2) \cot 2 - \frac{(x-2)^2}{2} \cos$$

$$+ \frac{(x-2)^3}{6 3} \operatorname{cosec}^2(2) \cot(2) + \ldots$$

(5) Find the expansion of $\tan\left(x + \frac{\pi}{4}\right)$ in ascendin powers of x upto terms in x^4 and find approximately the value of $\tan(43^\circ)$.

Let, $f\left(x + \frac{\pi}{4}\right) = \tan\left(x + \frac{\pi}{4}\right)$

$\therefore f(x) = \tan x$.

By Taylor's series,

$$f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \frac{h^4}{4!} f^{iv}(x)$$

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Putting $x = \frac{\pi}{4}$, $h = x$

$$f\left(\frac{\pi}{4}+x\right) = f\left(\frac{\pi}{4}\right) + x\cdot f'\left(\frac{\pi}{4}\right) + \frac{x^2}{2!} f''\left(\frac{\pi}{4}\right) + \frac{x^3}{3!} f'''\left(\frac{\pi}{4}\right) + \frac{x^4}{4!} f^{IV}\left(\frac{\pi}{4}\right) + \cdots \quad (1)$$

$f(x) = \tan x$, $f\left(\frac{\pi}{4}\right) = \tan\frac{\pi}{4} = 1$

$f'(x) = \sec^2 x$, $f'\left(\frac{\pi}{4}\right) = \sec^2\frac{\pi}{4} = 2$ $\{1+\tan^2 x = \sec^2 x$

$= 1+1 = 2$

$f''(x) = 2\sec x \cdot \sec x \cdot \tan x$, $f''\left(\frac{\pi}{4}\right) = 2\cdot\tan\left(\frac{\pi}{4}\right) + 2\tan^3\left(\frac{\pi}{4}\right)$

$= 2\sec^2 x \cdot \tan x$

$= 2(1+\tan^2 x)\cdot\tan x$

$= 2\tan x + 2\tan^3 x$

$= 2\cdot(1) + 2(1)$

$= 4$ and so on.

Putting eq^n (1),

$f''(x) = 2[\sec^2 x \sec^2 x] + 2\sec x \cdot \sec x \cdot \tan x$

$f'''(x) = 2\sec^2 x + 6\tan^2 x \cdot \sec^2 x$., $f'''\left(\frac{\pi}{4}\right) = 2\cdot\sec^2\left(\frac{\pi}{4}\right) + 6\tan^2\left(\frac{\pi}{4}\right)\sec^2\left(\frac{\pi}{4}\right)$

$= 2(1+\tan^2 x) + 6\tan^2 x(1+\tan^2 x)$

$= 2 + 2\tan^2 x + 6\tan^2 x + 6\tan^4 x$

$= 2 + 8\tan^2 x + 6\tan^4 x$

$= 2(2) + 6(2)$

$= 16$

~~$f^{IV}(x) = 4\sec x \cdot \{\sec x \cdot \tan x\}$~~

$f^{iv}(x) = 16\tan x \cdot \sec^2 x + 24\tan^3 x \cdot \sec^2 x$

$= 16\tan x(1+\tan^2 x) + 24\tan^3 x(1+\tan^2 x)$

$= 16\tan x + 16\tan^3 x + 24\tan^3 x + 24\tan^5 x$

$\therefore f^{iv}\left(\frac{\pi}{4}\right) = 16\tan\left(\frac{\pi}{4}\right) + 16\tan^3\left(\frac{\pi}{2}\right) + 24\tan^3\left(\frac{\pi}{4}\right) + 24\tan^5\left(\frac{\pi}{4}\right)$

$= 16 + 16 + 24 + 24$

$= 80$. and so on.

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∴ From eq^{n} (1),

$$f\left(\frac{\pi}{4}+x\right) = 1 + x\cdot(2) + \frac{x^2}{2!}(4) + \frac{x^3}{3!}(16) + \frac{x^4}{4!}(80) + \ldots$$

$$\tan\left(\frac{\pi}{4}+x\right) = 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \frac{10}{3}x^4 + \cdots$$

Now, $\tan 43^\circ = \tan(45^\circ - 2^\circ)$ $\quad \{1^\circ = \frac{\pi}{180}$ radian

$$= \tan\left(\frac{\pi}{4} - \frac{2\pi}{180}\right)$$

$$= \tan\left(\frac{\pi}{4} - 0.0349\right)$$

$$= 1 + 2(-0.0349) + 2(-0.0349)^2 + \frac{8}{3}(-0.0349)^3 + \frac{10}{3}(-0.0349)^4$$

$$= 0.9326 \quad \text{Approx.}$$

⑥ Using Taylor's thm., evaluate upto the 4 places of decimal. : $\sqrt{36.12}$

Let, $f(x) = \sqrt{x}$

$f(x+h) = \sqrt{x+h}$

By Taylor's series,

$$f(x+h) = f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots \quad (1)$$

To find $\sqrt{36.12}$

Putting $x = 36$, $h = 0.12$ in eq^{n} (1),

$$f(x+h) = \sqrt{x+h} = \sqrt{36+0.12} = \sqrt{36.12}$$

$$= f(36) + (0.12)f'(36) + \frac{(0.12)^2}{2!}f''(36) + \cdots$$

$f(x) = \sqrt{x}$, $\quad f(36) = \sqrt{36} = 6$

$f'(x) = \frac{1}{2\sqrt{x}}$, $\quad f'(36) = \frac{1}{2\sqrt{36}} = \frac{1}{12}$

$f''(x) = \frac{-1}{4x^{3/2}}$, $\quad f''(36) = \frac{-1}{4(36)^{3/2}} = \frac{-1}{4(6)^3} = \frac{-1}{864}$

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Substituting in eq^{n} (1),

$$\sqrt{36.12} = 6 + (0.12)\left(\frac{1}{12}\right) + \frac{(0.12)^2}{2!}\left(-\frac{1}{864}\right) + \cdots$$

$$= 6.0099 \text{ approx.}$$

Maclaurin's Series:

If $f(x)$ is a given function of x which can be expanded in positive ascending integral powers of x then

$$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots + \frac{x^n}{n!} f^n(0) + \cdots$$

This series can also be written as

$$y = y(0) + x \cdot y_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0) + \cdots + \frac{x^n}{n!} y_n(0) + \cdots$$

Standard Expansions:-

With the help of Maclaurin's series, we can get some standard expansions. This expansions can be used in solving examples.

1) Expansion of e^x (exponential series):

Let, $y = e^x$

$$y(0) = e^0 = 1$$

Now, $y_1(0) = 1$, $y_2(0) = 1$, $\cdots$, $y_n = \frac{d^n(e^x)}{dx^n} = e^x$

$$\therefore y_n(0) = e^0 = 1$$

Substituting in Maclaurin's series,

$$e^x = 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(1) + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This series is known as exponential series.

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2) Expansion of $\sin x$ (Sine series)

Let, $y = \sin x$

$y(0) = \sin 0 = 0$

Now, $y_n = \frac{d^n}{dx^n}(\sin x) = \sin\left(x + n\frac{\pi}{2}\right)$

$y_n(0) = \sin\left(\frac{n\pi}{2}\right)$

Putting $n = 1, 2, 3, 4, 5, \ldots$

$y_1(0) = \sin\frac{\pi}{2} = 1$, $y_2(0) = \sin\pi = 0$

$y_3(0) = \sin\frac{3\pi}{2} = -1$, $y_4(0) = \sin 2\pi = 0$

$y_5(0) = \sin\frac{5\pi}{2} = 1$ and so on.

Substituting in Maclaurin's series,

$\sin x = x - \frac{x^3}{3!}0 + x\cdot(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(0) + \frac{x}{5}$

and so on.

$\therefore \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$

3) Expansion of $\cos x$ (Cosine series)

Let, $y = \cos x$

$y(0) = \cos 0 = 1$

Now, $y_n = \frac{d^n}{dx^n}(\cos x) = \cos\left(x + \frac{n\pi}{2}\right)$

$y_n(0) = \cos\frac{n\pi}{2}$

Putting $n = 1, 2, 3, 4, \ldots$

$y_1(0) = \cos\frac{\pi}{2} = 0$, $y_2(0) = \cos\pi = -1$

$y_3(0) = \cos\frac{3\pi}{2} = 0$, $y_4(0) = \cos 2\pi = 1$, and so on

$y_3(0) = \cos\frac{3\pi}{2} = \cos\left(\frac{\pi}{2} + \pi\right) = \sin\pi = 0$

$y_4(0) = \cos 2\pi = 1$

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Substituting in Maclaurin's series,

$$\cos x = 1 + x(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(1) + \cdots$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

This series is known as cosine series.

) Expansion of $\tan x$

Let, $y = \tan x$, $y(0) = 0$

$$y_1 = \sec^2 x$$

$$= 1 + \tan^2 x,$$

$$= 1 + y^2 \quad , \quad y_1(0) = 1 + 0 = 1$$

$$y_2 = 2yy_1 \quad , \quad y_2(0) = 2y(0)y_1(0) = 2(0)(1) = 0$$

$$y_3 = 2y_1^2 + 2yy_2 \, , \, y_3(0) = 2(1)^2 + 2(0)(0) = 2$$

$$y_4 = 4y_1y_2 + 2y_1y_2 + 2yy_3 \, , \, y_4(0) = 4y_1(0)y_2(0) + 2y_1(0)y_2(0) + 2y(0)y_3(0)$$

$$= 4(0) + 2(1)(0) + 2(0)$$

$$= 0$$

$$y_5 = 4y_2^2 + 4y_1y_3 + 2y_2^2 + 2y_1y_3 + 2y_1y_3 + 2yy_4$$

$$= 0 + 4(1)(2) + 2(0) + 2(1)(2) + 2(1)(2) + 2(0)$$

$$= 8 + 4 + 4 = 16$$

Substituting in Maclaurin's series,

$$\tan x = y(0) + \frac{x}{1!}y_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0)$$

$$= 0 + \frac{x}{1!}(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(16)$$

$$= x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$$

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OR

$$\tan x = \frac{\sin x}{\cos x}$$

$$\tan x = \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots}{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots}$$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left(\frac{1}{1} - \frac{2!}{x^2} + \frac{4!}{x^4} - \cdots\right)$$

$$= x - \frac{2!}{x} + \frac{4!}{x^3} - \frac{x^3}{3!} + \frac{2!\,x^3}{3!\,x^2} - \frac{4!\,x^3}{3!\,x^4} + \frac{x^5}{5!} - \frac{x^5\,2!}{5!\,x}$$

$$+ \frac{x^5\,4!}{5!\,x^4} - \cdots$$

$$= x - \frac{24}{x} + \frac{24}{x^3} - \frac{x^3}{3!} + \frac{x}{3} - \frac{4}{x} + \frac{x^5}{5!} - \frac{x^3}{6}$$

$$+ \frac{x}{5}$$

$$= x - \frac{6}{x}$$

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